Mal de Débarquement Syndrome (MdDS): A Comprehensive Disease Characteristics Report

Disease Name: Mal de Débarquement Syndrome (MdDS) **Category:** Acquired **MONDO:** "mal de debarquement" (acquired disease term exists in Mondo); **MeSH:** "Mal de Debarquement" (introduced 2018); **ICD-11:** foundation term under vestibular/balance disorders; **ICD-10:** no dedicated code (coded under H81 / R42 "Dizziness and giddiness"); **OMIM:** none (non-Mendelian); **Orphanet:** listed among rare vestibular disorders.

Summary

Mal de Débarquement Syndrome (MdDS) is a rare, **acquired chronic central/functional vestibular disorder** defined by a persistent, non-spinning oscillatory perception of self-motion — described by patients as *rocking, bobbing, or swaying* — that is present continuously or for most of the day. The hallmark that distinguishes it from other chronic dizziness syndromes is its temporal relationship to passive motion: symptoms begin **within 48 hours of ceasing prolonged passive motion** (classically a sea cruise, but also air and road travel) and are **paradoxically relieved by re-exposure to passive motion** (e.g., driving). A non-motion-triggered ("spontaneous") variant also exists, overlapping clinically with persistent postural-perceptual dizziness (PPPD). These features are codified in the 2020 Bárány Society Classification Committee consensus criteria (PMID: 32986636).

Mechanistically, converging evidence supports a model of **maladaptive neuroplasticity** rather than any structural inner-ear lesion or gene defect. The leading physiological account holds that MdDS results from maladaptation of the **vestibulo-ocular reflex (VOR)** and the central **velocity-storage mechanism** to roll of the head during rotation — an inappropriate conditioning of a cross-axis coupling within a dynamical velocity-storage system (PMID: 25076935; PMID: 41122084). Neuroimaging shows a self-sustaining cortico-limbic network signature: **hypermetabolism of the left entorhinal cortex and amygdala** with altered

functional connectivity to posterior sensory-processing and frontal/temporal regions (PMID: 23209584; PMID: 33746890), and high-density EEG frames the condition as a brain state of **entrainment to oscillating motion** (PMID: 30099627).

MdDS predominates in **midlife women** (~3:1 to >4:1 female), is **highly comorbid with migraine**, is diagnosed **clinically by exclusion** (normal vestibular test battery and structural MRI), and is not life-threatening — but it is frequently chronic and disabling. The most disease-specific therapy is **VOR readaptation / optokinetic stimulation** (~70% substantial improvement in the original series), supplemented by neuromodulation (rTMS/tDCS over the dorsolateral prefrontal cortex) and migraine-prophylaxis pharmacotherapy. There is no FDA-approved drug, no causal gene, no animal disease model, and no established primary prevention. This report synthesizes 16 confirmed findings across 30 reviewed papers into a complete disease-characteristics entry.

Key Findings

1. Definition and Diagnostic Criteria (F001)

MdDS is a chronic vestibular disorder of persistent oscillatory self-motion. The Bárány Society Classification Committee consensus (PMID: 32986636) provides the authoritative case definition. The criteria specify: *"1] Non-spinning vertigo characterized by an oscillatory perception ('rocking,' 'bobbing,' or 'swaying') present continuously or for most of the day; 2] Onset occurs within 48 hours after the end of exposure to passive motion, 3] Symptoms temporarily reduce with exposure to passive motion (e.g. driving), and 4] Symptoms persist for >48 hours."*

Temporal qualifiers structure the diagnosis: **"in evolution"** (<1 month of observation), **"transient"** (resolves within ≤1 month), and **"persistent"** (>1 month). A **non-motion-triggered variant** is recognized, which follows another vestibular disorder, medical illness, psychological stress, or metabolic disturbance and overlaps clinically with PPPD. MdDS is classified within the International Classification of Vestibular Disorders (ICVD) as a chronic functional/central vestibular disorder.

2. Pathophysiology — VOR / Velocity-Storage Maladaptation (F002)

The mechanistic cornerstone is the proposal by Dai, Cohen and colleagues that MdDS arises from **maladaptation of the VOR to roll of the head during rotation**, derived from both monkey and human data: *"Results in monkeys and humans suggested that MdDS was caused by maladaptation of the vestibulo-ocular reflex (VOR) to roll of the head during rotation"* (PMID: 25076935). In a cohort of 24 subjects, physical findings included **body oscillation at ~0.2 Hz**, **oscillating vertical nystagmus on side-to-side head roll in darkness**, and unilateral rotation on the Fukuda stepping test.

More recent computational modelling formalizes the central **velocity-storage mechanism** as a 3×3 dynamical system: *"A central vestibular neural mechanism known as velocity storage may be inappropriately conditioned in mal de débarquement syndrome (MdDS)"* (PMID: 41122084). In this framework, maladapted off-diagonal (cross-axis coupling) elements — a misalignment between the yaw eigenvector and the head-vertical/gravity axis — produce the persistent "pull" / rocking sensation. This is elaborated as *"improperly sustained neuroplasticity in the velocity storage mechanism of the central vestibular system"* (PMID: 42440785).

3. Neuroimaging — Limbic Hypermetabolism and Altered Connectivity (F003)

Cha et al. studied 20 MdDS subjects (median duration 17.5 months) versus 20 controls using FDG-PET and resting-state fMRI, reporting: *"MdDS subjects showed increased metabolism in the left entorhinal cortex and amygdala (z>3.3)"* (PMID: 23209584). The same study found relative hypometabolism in the left superior medial/middle frontal gyri, right amygdala, right insula, and temporal gyri, alongside increased connectivity between the entorhinal/amygdala cluster and posterior visual/vestibular processing areas.

A dedicated review synthesizes these imaging findings (PMID: 33746890): *"a limbic focus in the left entorhinal cortex and amygdala may be important in the pathology of MdDS, as these structures are hypermetabolic in MdDS and exhibit increased functional connectivity to posterior sensory processing areas and reduced connectivity to the frontal and temporal cortices."* Voxel-based morphometry additionally shows **decreasing anterior cingulate volume** and **increasing inferior frontal gyri / anterior insula volume** with longer illness duration.

4. Mechanism as an Oscillatory-Entrainment Brain State (F014)

High-density resting-state EEG frames MdDS as *"a motion perceptual disorder induced by entrainment to oscillating motion"* (PMID: 30099627). In 20 women (mean age 52.9 ± 12.6 y; illness duration 35.2 ± 24.2 mo), rTMS-induced symptom improvement correlated with

increased long-range low-alpha (8–10 Hz) inter-regional phase coherence and decreased coherence in other bands, mostly between frontal and parietal regions. High baseline high-alpha/beta coherence predicted treatment response. This electrophysiological signature complements the FDG-PET/fMRI evidence and positions the disorder as a network-level dysrhythmia rather than a focal lesion.

5. Treatment — VOR Readaptation and Optokinetic Stimulation (F004)

The most disease-specific therapy directly targets the proposed velocity-storage maladaptation. Dai et al. treated 24 MdDS subjects by rolling the head side-to-side while viewing a rotating full-field visual stimulus: *"Seventeen of the 24 subjects had a complete or substantial recovery on average for approximately 1 year"* — approximately **70% response** (PMID: 25076935); 6 relapsed and 1 was a non-responder. The authors summarize that *"readaptation of the VOR has led to a cure or substantial improvement in 70% of the subjects with MdDS."*

The approach has been extended to sham-controlled optokinetic stimulation trials (PMID: 30410464) and translated to audiology-vestibular clinic settings. A case report of a 48-year-old woman treated with the **"Roll Readaptation"** technique — full-field omnidirectional optokinetic stimulus during rhythmic head roll, three short sessions — produced significant symptom reduction and return to full-time work after nearly 3 months off (PMID: 36323329).

6. Neuromodulation — rTMS, tDCS, iTBS over DLPFC (F010)

Controlled-trial evidence supports neuromodulation as an adjunctive therapy targeting the cortico-limbic network node:

Study	Design	N	Intervention	Key result
Cha et al. (PMID: 27176615)	Double-blind sham-controlled crossover	8 women	5 days 10 Hz rTMS, left DLPFC	Improved DHI at post-weeks 1,3,4 ($p<0.05$); improved HADS anxiety/depression; no change with sham
Cha et al. (PMID: 27117283)	Single-blind sham RCT, home tDCS after rTMS	23	tDCS anode L / cathode R DLPFC	Improved MdDS Balance Rating Scale and anxiety by week 4; safe (0 skin burns / 556 sessions)
Browne et al. (PMID: 38345630)	RCT, iTBS + VOR rehab vs sham	20	iTBS adjunct to VOR rehabilitation	Both groups improved; **no between-group difference** — iTBS added no benefit over VOR rehab alone
Scoping review (PMID: 40228811)	Review, 7 studies	—	TMS for chronic vestibular disorders	Statistically significant DHI improvement in 3/7; postural control improved in 7/7

Cha et al. concluded: *"Our study provides evidence that the dizziness, mood and anxiety symptoms of MdDS can be improved with 10 Hz rTMS over left DLPFC beyond the treatment period in selected individuals"* (PMID: 27176615). However, the scoping review found: *"Statistically significant improvements were noted on the Dizziness Handicap Inventory (3/7 studies) but clinically significant improvements were not observed"* (PMID: 40228811). rTMS response also has a connectivity correlate: improvement correlated with reduced connectivity between left entorhinal cortex and posterior default-mode nodes, and higher baseline DLPFC–entorhinal connectivity predicted response (PMID: 28967282).

7. Treatment — Vestibular-Migraine Prophylaxis (F009)

Given the high migraine comorbidity, migraine-prophylactic pharmacotherapy benefits many patients. Ghavami et al. treated 15 MdDS patients (73% female, mean age 50 ± 13 y) with an institutional vestibular-migraine protocol: *"Eleven patients (73%) responded well to management with a vestibular migraine protocol, which included lifestyle changes, as well as pharmacotherapy with verapamil, nortriptyline, topiramate, or a combination thereof"* (PMID: 27730651). Nearly all had a personal or family migraine history, and the response rate exceeded that of a retrospective vestibular-rehabilitation control group. Chronic-dizziness management additionally emphasizes serotonergic antidepressants that *"modulate sensory gating and reduce anxiety,"* vestibular rehabilitation, CBT, and trigger avoidance (PMID: 34351113). Benzodiazepines (e.g., clonazepam) are used symptomatically. **There is no FDA-approved drug specifically for MdDS.**

8. Epidemiology and Demographics (F005, F015)

MdDS shows a **strong female predominance** (21/24 female in the Dai cohort; 73% female in Ghavami's series) with typical **adult onset in the 4th–6th decades** (mean ages across cohorts: 44.5 ± 7.0, 50 ± 13, 52.9 ± 12.6 years). A sex-hormone review notes: *"In females, gonadal hormones and sex-specific synaptic plasticity may play a significant role in the underlying pathophysiology of peripheral and central vestibular disorders"* (PMID: 34864753), consistent with frequent perimenopausal onset.

Migraine comorbidity is high: in a study of rocking dizziness, *"both groups had a comparable prevalence of migraine headache (41%: MT; 46%: non-MT)"* (PMID: 23674832). True population prevalence and incidence are **unknown** — MdDS is considered rare and under-recognized, with no registry-based figures. Information derives from disease-level case series and specialty-clinic cohorts rather than population EHR data. No specific ethnic or geographic clustering is established.

Synonyms/nomenclature: Mal de débarquement syndrome; "sickness of disembarkment / disembarkment syndrome"; "debarkment syndrome"; MdDS; historically "landsickness."

9. Diagnosis by Exclusion (F006)

MdDS is diagnosed clinically per Bárány criteria; **no confirmatory laboratory test or biomarker exists.** Standard vestibular function testing (VNG/ENG, caloric, rotary chair, VEMP), audiometry, and structural brain MRI are characteristically normal: *"We found normal inner-ear function, non-related abnormalities and normal brain imaging"* (PMID: 32364688). FDG-PET limbic hypermetabolism and resting-state fMRI/EEG connectivity changes are **research-only** biomarkers. The one described diagnostic physical sign is oscillating vertical nystagmus on head roll in darkness (PMID: 25076935).

Key differential diagnoses include *"persistent postural perceptual dizziness, mal de débarquement syndrome, motion sickness and visually induced motion sickness, bilateral vestibulopathy"* (PMID: 34351113), plus vestibular migraine, BPPV, and Ménière's disease. The **distinguishing feature is transient relief with re-exposure to passive motion.**

10. Natural History and Prognosis (F007)

MdDS symptoms characteristically persist for months to years (median 17.5 months in the imaging cohort; 19.1 ± 33 months in the treatment cohort). Ghavami et al. note a key prognostic threshold: *"symptoms that persist beyond 6 months have been described as unlikely to remit"* (PMID: 27730651). The course is **fluctuating/relapsing**, exacerbated by psychological stress, fatigue, hormonal changes, and busy visual environments; Cha describes these as *"chronic syndromes with fluctuations that are both innate and driven by environmental stressors"* (PMID: 34351113). **MdDS is not associated with increased mortality;** morbidity arises from disability, occupational impairment, anxiety/depression, and reduced quality of life (PMID: 37987715; PMID: 40296474).

11. Etiology and Environmental Triggers (F011)

The defining environmental cause is **prolonged exposure to passive oscillatory motion** — classically sea travel, also air and road travel — with onset within 48 hours of disembarking: *"Onset occurs within 48 hours after the end of exposure to passive motion"* (PMID: 32986636). Occupational exposure is documented in pilots, where *"As flight time and age increased, the severity of the symptoms of MdDS increased for all subfactors"* (PMID: 40296474), and in military personnel exposed to transport motion (PMID: 37987715). Symptom-exacerbating factors include busy visual environments, fatigue, sleep deprivation, psychological stress, and hormonal fluctuations. **No toxin, radiation, pollutant, drug, or infectious agent** is implicated. Paradoxically, re-exposure to passive motion transiently relieves symptoms.

12. Phenotype Spectrum (F012)

The core phenotype (100% by definition) is **continuous non-spinning oscillatory self-motion** (rocking, bobbing, swaying) present most of the day, adult-onset, chronic/fluctuating. Frequently co-occurring symptoms: *"Individuals with MdDS may develop co-existing symptoms of spatial disorientation, visual motion intolerance, fatigue, and exacerbation of headaches or anxiety"* (PMID: 32986636). The symptom burden is variable across patients: even *"dizziness, fatigue, and brain fog, were endorsed variably across subjects"* (PMID: 42440785). Imbalance is largely subjective, though objective postural sway at ~0.2 Hz and oscillating vertical nystagmus on head roll are described. Quality-of-life impact is significant, including occupational disability.

Suggested HPO terms: Vertigo (HP:0002321), Abnormal vestibular function (HP:0410008), Fatigue (HP:0012378), Anxiety (HP:0000739), Depressivity (HP:0000716), Impaired concentration / cognitive impairment (HP:0100543), Nystagmus (HP:0000639).

13. Genetic/Molecular Basis — Non-Mendelian (F008)

No causal gene, pathogenic variant, chromosomal abnormality, or Mendelian inheritance pattern has been identified. MdDS is not catalogued in OMIM as a gene-associated disorder; it is an **acquired, multifactorial functional/central vestibular disorder** triggered by environmental motion exposure (PMID: 32986636; PMID: 33746890). Proposed molecular-level contributors are **neuromodulatory/hormonal rather than genetic**: gonadal (estrogen) influence on vestibular synaptic plasticity (PMID: 34864753) and migraine-related physiology (CGRP, serotonergic/sensory-gating pathways) given high migraine comorbidity. No transcriptomic, proteomic, metabolomic, or epigenetic disease signature has been established. Germline/somatic variants, modifier genes, founder effects, penetrance, and carrier frequencies are **not applicable**.

14. Prevention, Other Species, and Model Organisms (F013)

Prevention: No vaccine, screening program, or proven primary prevention exists. Practical risk reduction is behavioral — limiting/preparing for prolonged provocative passive motion, and, once symptomatic, avoiding triggers, managing stress/sleep, and initiating early VOR-readaptation therapy (tertiary prevention of chronicity). Persistence >6 months predicts lower remission, arguing for early intervention (PMID: 27730651; PMID: 34351113). Genetic counseling is not applicable.

Other species / natural disease: MdDS is a **human-specific clinical entity**; no naturally occurring MdDS is documented in companion animals or wildlife (no OMIA entry).

Model organisms: There is **no transgenic/knockout genetic model.** The mechanistic underpinning — velocity storage and roll-while-rotating VOR adaptation — was characterized experimentally in **non-human primates (Macaca; NCBI Taxon 9544)** and modeled computationally, informing the human treatment (PMID: 25076935; PMID: 41122084). Rotating optokinetic/roll paradigms in humans serve as the principal experimental system.

15. Integrated Synthesis — A Treatable Maladaptive-Plasticity Network Disorder (F016)

Convergent evidence supports a unified causal model: prolonged passive oscillatory motion entrains the central velocity-storage/VOR mechanism, producing a self-sustaining cortico-limbic network dysrhythmia (left entorhinal/amygdala hypermetabolism; altered default-mode/salience/executive and fronto-parietal coherence), which manifests as chronic internal rocking with fatigue, brain fog, visual-motion intolerance, and anxiety (PMID: 23209584; PMID:

33746890; PMID: 30099627; PMID: 41122084). Treatment targets each node — VOR/optokinetic readaptation (velocity storage; ~70% response) and DLPFC/cerebellar neuromodulation (network). Response is **heterogeneous and increasingly personalized**: two velocity-storage strategies (correction vs. attenuation) yield differing outcomes, and visual-motion sensitivity predicts poorer response to attenuation approaches (PMID: 42440785; PMID: 41122084).

Mechanistic Model / Interpretation

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  TRIGGER (upstream)              CENTRAL MALADAPTATION            NETWORK STATE (downstream)
+--------------------+      +-----------------------------+    +----------------------------+    -
| Prolonged passive  |      | Velocity-storage / VOR      |    | Cortico-limbic dysrhythmia|
| oscillatory motion | ---> | maladaptation:              |--> | - L entorhinal cortex +   |-->
| (cruise, flight,   |      | roll-while-rotating         |    |   amygdala HYPERmetabolism|
| car); onset <48 h  |      | cross-axis coupling         |    | - altered DMN/salience/   |
| after motion ends  |      | (3x3 dynamical system)      |    |   fronto-parietal coherence|
+--------------------+      +-----------------------------+    +----------------------------+    -
       |                              ^                                  ^
       |  re-exposure to motion       |  VOR / optokinetic               |  rTMS / tDCS over DL
       +--- transiently RELIEVES -----+  READAPTATION (~70%)             +- neuromodulation (ac
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Modifiers/amplifiers: female sex and gonadal hormones (perimenopausal onset), migraine physiology (CGRP, serotonergic sensory gating), psychological stress, fatigue, sleep deprivation, and busy visual environments. The paradoxical relief on re-exposure to motion is a defining clue that the disorder reflects a *learned/entrained internal model* that is transiently "matched" when real motion resumes.

Ontology term suggestions: - **UBERON / anatomy:** vestibular system, semicircular canal (UBERON:0001840), brainstem vestibular nuclei, entorhinal cortex (UBERON:0002728), amygdala (UBERON:0001876), dorsolateral prefrontal cortex, cerebellum (UBERON:0002037), insula. - **CL / cell types:** central vestibular neurons; no specific pathological cell population is identified (no cell death or lesion). - **GO / biological process:** vestibulo-ocular reflex (GO:0060013), regulation of neuronal synaptic plasticity (GO:0048168), adaptation of signaling pathway, sensory perception of balance. - **CHEBI / chemicals (therapeutic):** verapamil, nortriptyline, topiramate, clonazepam; estradiol/estrogen (modifier). - **NCIT / interventions:** vestibular rehabilitation therapy, transcranial magnetic stimulation, transcranial direct current stimulation, cognitive behavioral therapy. - **HPO:** see Finding 12.

Evidence Base

PMID	Focus	Contribution
32986636	Bárány Society diagnostic criteria	Authoritative case definition, temporal qualifiers, triggers, ICVD classification
25076935	VOR readaptation relieves MdDS	Core mechanism (VOR maladaptation) + landmark ~70%-response treatment; NHP + human data
41122084	Model-based treatment-effect heterogeneity	Velocity-storage 3x3 dynamical model; personalization rationale
42440785	Toward personalized medicine for MdDS	Maladaptive-plasticity framing; variable symptom burden; correction vs. attenuation strategies
23209584	Metabolic/connectivity changes	Primary FDG-PET/fMRI evidence of limbic hypermetabolism
33746890	Neuroimaging markers review	Synthesis of limbic focus + connectivity/VBM changes
30099627	EEG signatures of rTMS treatment	Entrainment brain-state framing; EEG coherence biomarker; midlife-female demographics
28967282	RSFC signature of rTMS	Connectivity predictor/biomarker of rTMS response
27176615	Double-blind sham rTMS crossover	Controlled evidence rTMS improves dizziness/mood/anxiety
27117283	Home tDCS after rTMS RCT	tDCS extends benefit; home safety (0/556 burns)
38345630	iTBS + VOR rehab RCT	Negative: iTBS adds no benefit over VOR rehab
40228811	TMS scoping review	Modest / statistically-but-not-clinically-significant benefit
30410464	Sham-controlled optokinetic stimuli	Controlled support for optokinetic treatment
36323329	Roll Readaptation case (audiology)	Translation of readaptation to clinic; functional recovery
27730651	MdDS as vestibular migraine	73% response to migraine prophylaxis; >6-month prognosis threshold
23674832	Rocking dizziness & headache	Quantifies migraine comorbidity (41–46%)

PMID	Focus	Contribution
34864753	Sex hormones & vestibular disorders	Hormonal/plasticity basis; female predominance
34351113	Chronic Dizziness	Differential diagnosis; fluctuating course; management principles
32364688	"Sickness of disembarkment" review	Normal testing -> exclusion diagnosis
31580016	The MdDS (review)	Phenotype, female predominance, QoL, normal work-up
40296474	MdDS in pilots by flight time	Occupational exposure-response relationship
37987715	MdDS in military operations	Occupational burden; disability/morbidity

Convergence vs. challenge: The mechanistic (VOR/velocity-storage), imaging (limbic/network), and electrophysiological (entrainment) lines of evidence are mutually reinforcing. The main *challenge* within the treatment literature is the negative iTBS RCT (PMID: 38345630) and the scoping-review conclusion that TMS benefits are statistically but not clinically significant (PMID: 40228811) — tempering enthusiasm for neuromodulation as a stand-alone therapy and reinforcing VOR readaptation as the primary disease-specific intervention.

Limitations and Knowledge Gaps

1. **No population-based epidemiology.** True prevalence and incidence are unknown; all demographic estimates derive from small specialty-clinic case series (often N < 25), risking referral and sex-ascertainment bias.
2. **Small treatment trials.** The pivotal readaptation and neuromodulation studies enroll tens of patients; several are single-arm, crossover, or case reports. There are no large multicenter RCTs and no head-to-head comparisons of readaptation vs. neuromodulation vs. pharmacotherapy.
3. **No validated clinical biomarker.** FDG-PET, fMRI, and EEG signatures are research-only; diagnosis remains purely clinical and by exclusion, contributing to under-recognition and diagnostic delay.

4. **Mechanism is inferential.** The velocity-storage model is well-motivated by NHP physiology and computational modelling but not directly confirmed in human MdDS tissue or with causal manipulation; the relationship between the peripheral VOR account and the cortico-limbic imaging findings is correlational.
5. **No animal disease model.** Absence of a naturalistic or genetic model limits mechanistic dissection and therapeutic screening.
6. **Heterogeneous outcome measures.** Studies use DHI, MdDS Balance Rating Scale, HADS, and postural sway inconsistently, hindering meta-analysis.
7. **Non-motion-triggered variant is poorly delineated** from PPPD, blurring case boundaries.
8. **Molecular biology is essentially unexplored** — no transcriptomic, proteomic, metabolomic, or epigenetic studies; the hormonal/migraine hypotheses remain associative.

Proposed Follow-up Experiments / Actions

1. **Multicenter registry and prevalence study.** Establish a standardized MdDS registry (with the non-motion-triggered variant flagged) to derive population prevalence/incidence, sex ratio, and natural-history remission curves, formally testing the ">6-month = unlikely to remit" threshold.
2. **Definitive RCT of VOR/optokinetic readaptation** with sham control, standardized DHI/MdDS-BRS endpoints, and 12-month follow-up; pre-register correction vs. attenuation strategy arms stratified by baseline visual-motion sensitivity (the predictor identified in PMID: 42440785).
3. **Biomarker validation.** Prospectively test whether baseline DLPFC–entorhinal RSFC (PMID: 28967282) or EEG alpha/beta coherence (PMID: 30099627) predicts response, moving toward a treatment-selection tool.
4. **Hormonal mechanism study.** Characterize onset/severity relative to menstrual cycle, menopause, and hormone therapy to test the gonadal-hormone hypothesis (PMID: 34864753); consider a pilot of hormonal modulation.
5. **Migraine-overlap trial.** Head-to-head RCT of vestibular-migraine prophylaxis (verapamil/nortriptyline/topiramate) vs. readaptation vs. combination, given the 73% response signal (PMID: 27730651) and CGRP biology; explore anti-CGRP agents as a mechanistically motivated experimental therapy.

6. **Computational/personalized modelling.** Extend the velocity-storage dynamical model (PMID: 41122084) into a patient-specific readaptation-protocol optimizer, validated prospectively.
7. **Molecular pilot.** Exploratory plasma metabolomic/proteomic and, where feasible, blood transcriptomic profiling to seek any peripheral signature, acknowledging the low prior probability given the functional nature of the disorder.

Conclusion

Mal de Débarquement Syndrome is an acquired, non-genetic, chronic central/functional vestibular disorder in which prolonged passive oscillatory motion inappropriately conditions the brain's velocity-storage/VOR machinery, producing a self-sustaining cortico-limbic network state that is experienced as continuous internal rocking, bobbing, or swaying — accompanied by fatigue, brain fog, visual-motion intolerance, and anxiety. It predominates in midlife women, is highly comorbid with migraine, and is diagnosed clinically by exclusion in the presence of normal vestibular testing and MRI. The maladaptive-plasticity model is directly actionable: VOR readaptation / optokinetic therapy achieves ~70% substantial improvement, with adjunctive DLPFC neuromodulation and migraine-prophylaxis pharmacotherapy for selected patients. It is not life-threatening but is often chronic and disabling, and early intervention is favored because symptoms persisting beyond six months are less likely to remit spontaneously.
